

A Standard tumour sequencing report

TUMOUR SEQUENCING SUMMARY

Melanoma (SKCM) · Patient ID redacted

Intratumour heterogeneity

Subclonal fraction (h^2) 0.62

CCF variance (V_A) 0.140

Effective population size

N_e (VAF spectrum) 8,200 cells

Resistance loci (L) 10 loci

Actionable variants

BRAF p.V600E (clonal) 12.4 mut/Mb

B Resistance-learnability threshold

Computed from V_A , N_e , L

N_e^* threshold

$N_e^* = 9,262$ cells

$\varepsilon^* = 0.053$

Patient $N_e/N_e^* = 0.89$

BELOW N_e^*
resistance precluded

$$N_e^* = \min \{N_e : P(\text{failure}) \leq \delta\}$$
$$\varepsilon = 0.05, \delta = 0.05$$

*Clinical goal: keep N_e below N_e^**